

Jaluag, James

Kiah, Andre

Dr Choi

COSC 410

COSC 410 – Data Communications Project

Part 1 : Web Server Lab

File: *server.py*

Console

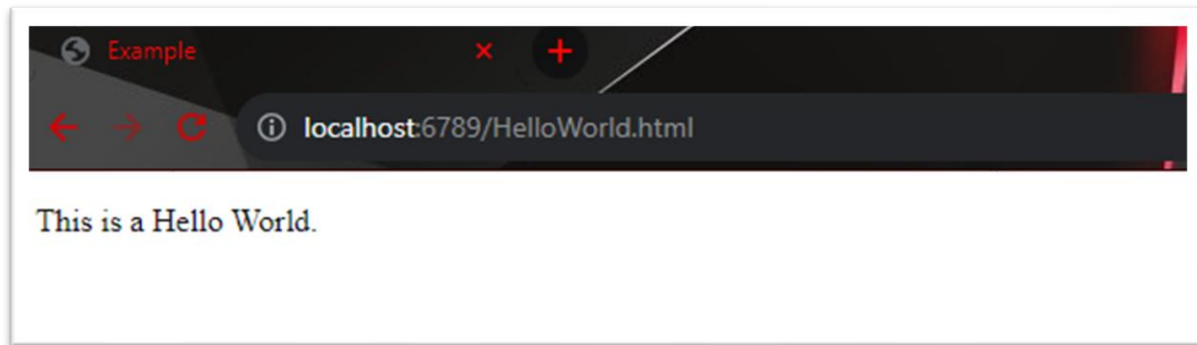
```
PS C:\Users\James\Desktop\COSC 410\Project> python -u "c:\Users\James\Desktop\COSC 410\Project\server.py"
Ready to serve...
```

Retrieval of HelloWorld.html file

```
PS C:\Users\James\Desktop\COSC 410\Project> python -u "c:\Users\James\Desktop\COSC 410\Project\server.py"
Ready to serve...
Message is: GET /HelloWorld.html HTTP/1.1
Host: localhost:6789
Connection: keep-alive
Cache-Control: max-age=0
sec-ch-ua: "Google Chrome";v="119", "Chromium";v="119", "Not?A_Brand";v="24"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Windows"
DNT: 1
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.9

File name is: /HelloWorld.html
OUTPUT DATA: ['<!DOCTYPE html>', '<html>', '    <head>', '        <title>Example</title>', '    </head>', '    <body>', '        <p>This is an Hello World.</p>', '    </body>', '</html>']
```

Web browser view

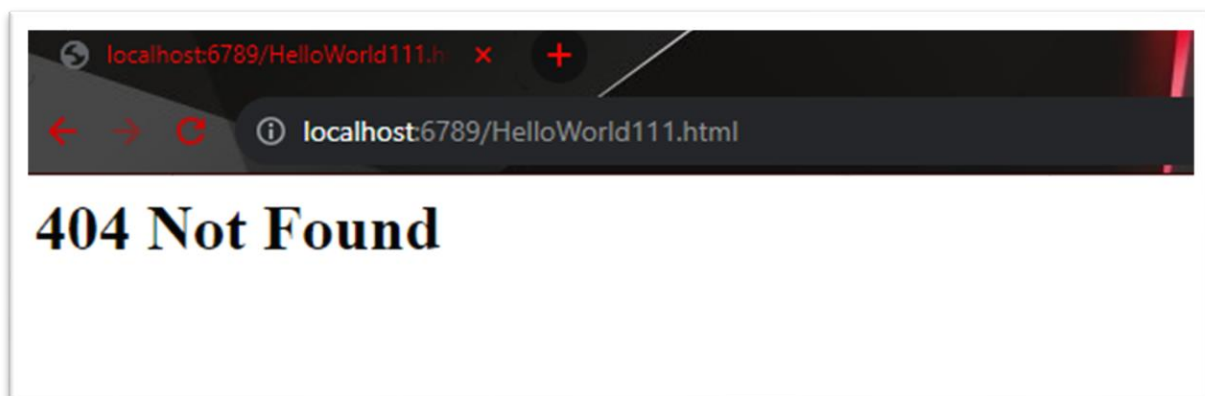


Entering a filename that does not exist in the server.

```
PS C:\Users\James\Desktop\COSC 410\Project> python -u "c:\Users\James\Desktop\COSC 410\Project\server.py"
Ready to serve...
Message is: GET /HelloWorld111.html HTTP/1.1
Host: localhost:6789
Connection: keep-alive
sec-ch-ua: "Google Chrome";v="119", "Chromium";v="119", "Not?A_Brand";v="24"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Windows"
DNT: 1
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Sec-Fetch-Site: none
Sec-Fetch-Mode: navigate
Sec-Fetch-User: ?1
Sec-Fetch-Dest: document
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.9

File name is: /HelloWorld111.html
```

Web browser view



Part 2 : Multithreaded Web Server

File: *multi2.py*

Consecutive file requests to the server

```
PS C:\Users\James\Desktop\COSC 410\Project> python -u "c:\Users\James\Desktop\COSC 410\Project\multi2.py"
Ready to serve...

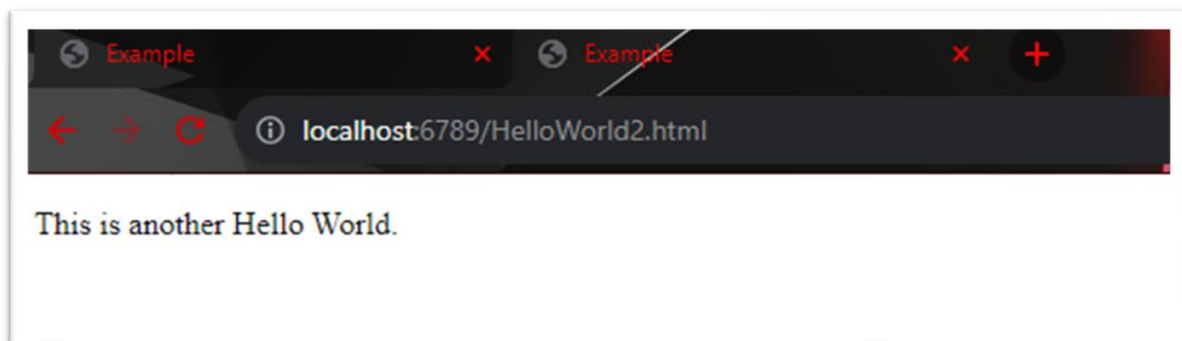
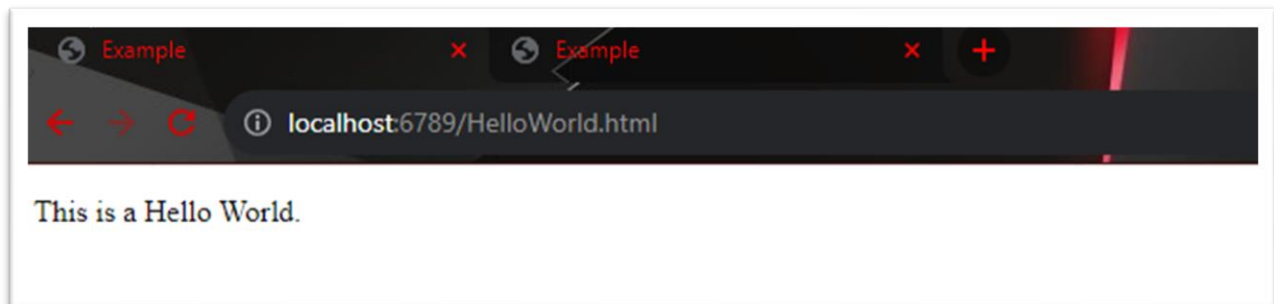
New connection added: ('127.0.0.1', 61300)
Ready to serve...
Message is: GET /HelloWorld.html HTTP/1.1
Host: localhost:6789
Connection: keep-alive
sec-ch-ua: "Google Chrome";v="119", "Chromium";v="119", "Not?A_Brand";v="24"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Windows"
DNT: 1
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Sec-Fetch-Site: none
Sec-Fetch-Mode: navigate
Sec-Fetch-User: ?1
Sec-Fetch-Dest: document
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.9

New connection added: File name is: /HelloWorld.html
('127.0.0.1', 61301)
Ready to serve...
OUTPUT DATA: ['<!DOCTYPE html>', '<html>', '    <head>', '        <title>Example</title>', '    </head>', '    <body>', '
    <p>This is an Hello World.</p>', '    </body>', '</html>']

New connection added: ('127.0.0.1', 61302)
Ready to serve...
Message is: GET /HelloWorld2.html HTTP/1.1
Host: localhost:6789
Connection: keep-alive
sec-ch-ua: "Google Chrome";v="119", "Chromium";v="119", "Not?A_Brand";v="24"
sec-ch-ua-mobile: ?0
sec-ch-ua-platform: "Windows"
DNT: 1
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Sec-Fetch-Site: none
Sec-Fetch-Mode: navigate
Sec-Fetch-User: ?1
Sec-Fetch-Dest: document
Accept-Encoding: gzip, deflate, br
Accept-Language: en-US,en;q=0.9

File name is: /HelloWorld2.html
OUTPUT DATA: ['<!DOCTYPE html>', '<html>', '    <head>', '        <title>Example</title>', '    </head>', '    <body>', '
    <p>This is another Hello World.</p>', '    </body>', '</html>']
```

Web browser view



Part 3 : Web Client

File: *client.py*

Running the client.py program and entering the arguments

```
PS C:\Users\James\Desktop\COSC 410\Project> python -u client.py localhost 6789 /HelloWorld.html
```

Receipt of the contents of HelloWorld.html file

```
PS C:\Users\James\Desktop\COSC 410\Project> python -u client.py localhost 6789 /HelloWorld.html
HTTP/1.1 200 OK
```

```
<!DOCTYPE html>
<html>
  <head>
    <title>Example</title>
  </head>
  <body>
    <p>This is an Hello World.</p>
  </body>
</html>
```

```
PS C:\Users\James\Desktop\COSC 410\Project> █
```